Weeks 1–3 Practice Artifacts — Fillable Backup

Use this document if Miro is unavailable or lagging. Fill in the tables below with your own Company A / Company B research, then paste the content into your group's Miro frame later, or use this document directly as part of your Assignment 1 appendix.

Week 1 — Environment & AI-Opportunity Map

Company selection

Company A — international / global comparator	Company B — Kazakhstan focal organization
Shell	KazMunayGas (KMG)

Name and Industry for each company — fill both cells above.

Environment Scavenger Hunt — 7 sectors × 2 companies

Sector	Company A — fact + source	Company B — fact + source
Economic	Oil and gas revenue is exposed to global commodity prices; Shell's integrated portfolio spans upstream, LNG, refining and marketing. Source: Shell Annual Report 2025, strategy/financial review.	KMG revenue and cash flow depend strongly on oil production, refining and domestic fuel sales; state ownership links performance to national energy priorities. Source: KMG Annual Report 2025.
Political / legal	Shell operates under UK listing, environmental, competition and climate-disclosure rules across many jurisdictions. Source: Shell Annual Report 2025, risk factors.	KMG operates under Kazakhstan's Subsoil and Subsoil Use Code, environmental rules and state regulation of some domestic fuel prices. Source: KMG Annual Report 2025; RK legal framework.
Demographic	Shell serves consumers and industrial customers across multiple countries and demographic markets.	KMG serves domestic motorists, airlines, agriculture, industrial users and international energy customers; demand varies by region and season.
Social	Shell faces public expectations on energy transition, emissions, affordability and responsible operations.	KMG has a strong social mandate around reliable domestic energy supply, employment, regional development and environmental impact.
Competitive	Shell competes with international oil majors, national oil companies and renewable-energy providers.	KMG competes with domestic and regional fuel suppliers and international producers; refinery capacity and logistics shape its position.
Global	Shell sells energy and trades commodities globally, with exposure to geopolitical and shipping disruptions.	KMG exports crude and petroleum products while balancing domestic supply obligations and access to export routes.
Technological	Shell uses digital operations, data analytics, automation and AI in trading, maintenance and customer operations.	KMG develops digitalisation and industrial-safety solutions, including TUMAR, and uses data/automation across production and refining.

KZ Snapshot

Indicator	Value	Source
GDP growth rate	5.0% (2024 actual; use latest official release if required)	Bureau of National Statistics of Kazakhstan, national accounts, 2024.
Inflation rate	8.6% (Dec. 2024 y/y; update to the assignment's reference year)	National Bank of Kazakhstan / Bureau of National Statistics, CPI release.
Unemployment rate	4.7% (2024; ILO/BNS measure)	Bureau of National Statistics of Kazakhstan, labour market statistics.

AI-Opportunity Map — one idea per sector, for Company B

Sector	AI opportunity idea for Company B
Economic	Use commodity-price, exchange-rate and demand forecasting to improve budgeting and investment scenarios.

Sector	AI opportunity idea for Company B
Political / legal	Use compliance analytics to monitor permits, emissions reporting and changing regulatory requirements; keep legal review human-led.
Demographic	Forecast workforce skills and retirement needs to plan technical recruitment and training.
Social	Analyse community feedback and service complaints to identify regional concerns; anonymise personal data.
Competitive	Use predictive maintenance and production analytics to reduce downtime and improve refinery competitiveness.
Global	Optimise export-route, shipping and inventory scenarios under geopolitical and logistics disruptions.
Technological	Expand predictive maintenance and industrial safety analytics, with engineer verification before safety-critical action.

Week 2 — Stakeholder & Ethics Map + Global Trade

Warm-up: The Bill Gannon dilemma — Board's decision

Bill Gannon, a middle manager, is pressured by his boss Dana Johnson to give employee Robert Talbot a positive review, despite genuine performance problems and a client complaint. A previous manager who gave Robert a bad review is no longer with the company. Discuss as a Board: (1) What ethical issues does this raise? (2) Should Bill confront Dana, or Dana's boss? (3) What would you do, and why? Reframe the dilemma for Company B and reach a Board decision.

Company B focus: KMG's AI opportunities should prioritise asset reliability, HSE, domestic fuel availability and compliance. Use AI for decision support; retain human accountability for safety, employment and legal decisions.

Write your Board's decision above, and note any unresolved tension.

Stakeholder & AI-Ethics Map

Category	Company A	Company B
Internal — employees: top concern?	Technical and field employees; concern: safe conditions, reliable equipment, skills and job security.	Production and refinery employees; concern: HSE, stable operations, training and safe digital change.
Internal — management priority?	Deliver shareholder value, disciplined capital allocation, reliable production and energy transition.	Balance profitability, production growth, national energy security, domestic supply and state/shareholder expectations.
Market — key customers?	Retail fuel customers, households, airlines, shipping, industrial and commercial energy buyers.	Domestic motorists, agriculture, transport fleets, airlines, industrial users and export customers.
Market — investors / suppliers?	Institutional shareholders, lenders, suppliers, trading partners and LNG/project contractors.	State shareholder, lenders, contractors, oilfield-service firms, pipeline/rail operators and joint-venture partners.
Societal — community / regulator concern?	Climate impact, methane/emissions, biodiversity, affordability and transition credibility.	Environmental impact, local communities near facilities, industrial safety, domestic fuel affordability and regulatory compliance.
AI-ethics flag (bias, privacy, jobs, data localization)?	AI may expose operational/customer data, reproduce biased forecasts or affect jobs; cyber risk is material to connected assets.	Protect employee and operational data; localise/manage data lawfully, test models across seasons/regions, and retain human review for safety-critical outputs.

Global Trade — Comparative Advantage & Trade Barriers

Factor	Company A	Company B
Market entry mode	Direct investment, subsidiaries, joint ventures and trading/marketing operations across countries.	Exports, joint ventures and subsidiaries; domestic operations remain central.
Main trade barrier faced	Host-country regulation, sanctions/geopolitical risk, carbon rules, tariffs and market-access requirements.	Export-route constraints, transit-country risk, sanctions compliance, product standards and domestic price regulation.
Comparative / absolute advantage	Global scale, trading capability, LNG portfolio, technology and diversified assets.	Access to Kazakhstan's hydrocarbon resources, established domestic infrastructure and regional market knowledge.
AI-related global risk or opportunity	AI can optimise trading, shipping and maintenance; risks include cyberattacks, model error and data-sovereignty constraints.	AI can improve predictive maintenance, logistics and HSE; opportunities depend on data quality, cybersecurity and local skills.

Transfer verdict — Company B's global-market move

Adopt	Adapt	Pilot	Delay	Reject
Predictive maintenance and digital asset monitoring, with human verification.	Shell's digital and customer practices to Kazakhstan's regulation, climate and domestic-supply role.	AI-supported fuel-demand and logistics forecasting at one refinery/region.	Fully autonomous safety-critical decisions until independently validated.	Dynamic surge pricing that conflicts with regulated domestic fuel pricing and affordability.

Circle or bold your verdict's column, and write your justification in that cell.

Week 3 — Ownership-Choice Case + AI Business Model Stress Test

Activity 1: Ownership-Choice Case

Criterion	Company A — current form	Company B — current form	Recommended form for Company B
Liability exposure	Limited liability; public shareholders' liability generally limited to their investment.	Joint-stock company; state is controlling shareholder, liability generally limited.	Retain joint-stock structure; it supports scale and limited liability.
Control and decision-making	Board and executive management operate under corporate governance; dispersed shareholders influence through voting.	State shareholder has significant influence; board and executive management run operations.	Strengthen board oversight and transparent delegation while preserving state strategic role.
Ability to raise capital	Can raise equity and debt through public markets and institutional investors.	Can raise debt and use state/strategic support; equity options are shaped by state ownership.	Use diversified debt/project finance and selective partnerships; preserve strategic control.
Continuity and transferability	Public company shares are transferable; continuity independent of individual owners.	Corporate continuity is stable; ownership transfer is constrained by state control.	Maintain JSC; improve minority-investor protections and governance.

Recommended ownership form for Company B

KMG should retain its joint-stock company form. It combines limited liability, continuity and access to capital while allowing the state to retain strategic control; governance transparency and board independence should be strengthened.

Write your recommendation and one-line justification above. Ground your answers in OpenStax Table 4.4.

Activity 2: AI Business Model Stress Test

Legend — Adopt: works as-is, implement directly. Adapt: core idea transfers, modify for local context. Pilot: promising but unproven, test small first. Delay: not ready yet, revisit after a trigger condition. Reject: doesn't fit Company B's context, resources or values.

Element	Question	Verdict	Justification
Value Proposition	What value does Company A offer? Match, adapt, or skip for Company B?	Adapt	Shell offers integrated, reliable energy solutions; KMG should tailor this to domestic fuel security, regional needs and Kazakhstan's energy transition.
Revenue Streams	How does Company A earn money? Transferable to Company B's market?	Adapt	Shell earns from upstream, LNG, refining, marketing and trading; KMG has similar streams but domestic price regulation and export logistics affect transferability.
Cost Structure	Which costs does AI change for Company A? Same for Company B?	Pilot	AI can reduce downtime and optimise trading/maintenance costs; KMG should pilot on selected assets and measure savings before scaling.
Customer Segments	Who does Company A serve? Same segments in Kazakhstan?	Adapt	Shell serves global consumers and industrial clients; KMG must prioritise Kazakhstan households, agriculture, transport and industrial customers alongside exports.

Element	Question	Verdict	Justification
Channels	How does Company A reach customers? Available for Company B?	Adapt	Shell uses retail, B2B, digital and trading channels; KMG should combine stations, rail/wholesale terminals, B2B contracts and digital services.
Key Resources and Partners	What makes Company A's model work? Does Company B have access?	Pilot	Shell's global technology, capital and supplier network is difficult to replicate fully; KMG should build local data skills and partner selectively.
Ownership and Legal Structure	Does Company A's ownership form enable this model? Fits Company B's context?	Adapt	Shell's public-company governance differs from KMG's state-controlled JSC; transfer practices only where consistent with Kazakhstan law and KMG's mandate.

Verdict = Adopt / Adapt / Pilot / Delay / Reject. Justification = one sentence.